



Pharmacy Management System

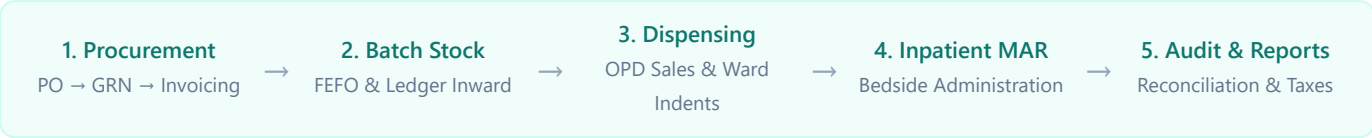
Complete End-to-End Operational Workflow & User Handbook

Authoritative workflow reference guide for hospital administrators, chief pharmacists, ward nurses, and billing cashiers. Covers the complete supply chain, atomic inventory ledger, point of sale, clinical ward indents, medication administration, and audit controls.

Hospital Facility:	Vatsalya Hospital Central Pharmacy
Technology Stack:	PHP 8.2+, MySQL 8/MariaDB, Vanilla JS, Bootstrap 5.3
Author / Maintainer:	WTS Systems & Healthcare Solutions
System Version:	1.0.0 Production Release
Web Application URL:	http://127.0.0.1:8000/

1. System Architecture & Role Access

The **Vatsalya Hospital Pharmacy Management System (PMS)** is an enterprise hospital pharmacy platform designed to maintain strictly atomic stock balances, eliminate medication dispensing errors, automate drug billing, and provide real-time clinical and financial oversight.



User Roles & Default Credentials

Role	Username	Default Password	Primary Responsibilities & Permissions
ADMIN	admin	Admin@12345	Full system control, user & role administration, audit logs, price overrides, stock write-offs, and hospital settings.
PHARMACIST	pharmacist	Pharma@12345	OPD billing, IPD ward indents, GRN physical verification, FEFO management, batch quarantine, and expiry reviews.
CASHIER	cashier	Cashier@12345	Fast-track OPD POS billing, receiving multi-mode payments (Cash, UPI, Card), and printing sales receipts.

Key Architectural Rule — Append-Only Stock Ledger: Stock quantities in this system are never arbitrarily overwritten. Every physical change (receiving delivery, selling at counter, ward dispensing, customer return, or disposal) writes an immutable row into the `pharmacy_stock_ledger` with previous balance, movement quantity, and new balance.

2. Outpatient (OPD) & Counter Sales Workflows

A. OPD Sales & POS Billing

Located at: `modules/sales/counter.php` (Sidebar: **OPD Sales**)

Engineered for high-volume retail billing of outpatient visitors and walk-in customers with automated FEFO batch allocation.

Step	User Action	System Backend Execution
1. Search Medicine	Type at least 2 characters of Brand Name or Composition in the search bar.	Live AJAX query searches active inventory. Filters out batches that are expired or quarantined. Displays live available stock and MRP.
2. Add to Cart	Click any medicine from search results. Specify quantity.	Validates quantity $\leq$ non-expired available stock. Automatically allocates earliest unexpired batches via FEFO . Pharmacist can click the FEFO preview button to see allocated batch numbers and dates.
3. Discounts & Taxes	Enter overall discount (%) or line-item discount.	Auto-computes taxable base, GST (0%, 5%, 12%, 18%), and grand total in real time.
4. Customer & Payment	Enter patient name, mobile, prescriber doctor, and select payment mode (Cash, UPI, Card, Bank Transfer, or Due/Credit).	Validates received amount. Automatically computes change due or balance outstanding.
5. Complete & Print	Click "Complete & Print Sale" button (includes double-click prevention).	Atomically acquires row-level locks, deducts batch quantities, writes SALE stock ledger entries, generates unique invoice (CS-000001), and renders printable thermal receipt with Vatsalya logo.

B. Regular / IPD Patient Sales

Located at: `modules/sales/regular.php` (Sidebar: **Regular / IPD Sale**)

Used when dispensing medication directly against an admitted Inpatient (IPD) bed. Links the sale to patient registration, admission ID, ward name, and bed number, permitting deferred credit billing until final discharge clearance.

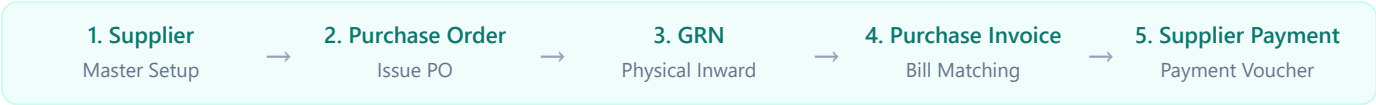
C. Sales Returns & Credit Notes

Located at: `modules/sales/returns.php` (Sidebar: **Sale Returns**)

Workflow Step	Controls & Safeguards
Invoice Lookup	Search by invoice number (e.g. CS-000001) or customer name. Loads all billed items and batch numbers.
Quantity Check	System verifies: $\text{Return Qty} \leq (\text{Billed Qty} - \text{Previously Returned Qty})$. Over-returns are strictly prohibited.
Restock vs Quarantine	Pharmacist selects condition: <i>Resellable / Good</i> (returned to active batch) or <i>Damaged / Compromised</i> (routed to Quarantine).
Financial Credit	Generates official Sales Return Note (SRT-000001) and adjusts cash drawer or customer credit due.

3. Procurement & Purchase Workflow (P2P)

The Procure-to-Pay (P2P) workflow governs all incoming stock from pharmaceutical manufacturers and wholesale distributors.



Stage	Module & Action	Business Rules & Ledger Impact
1. Supplier Registry	modules/purchases/suppliers.php Register distributor with GSTIN, Drug License (DL) numbers, contact info, and credit terms.	Enforces uniqueness of supplier codes and tax identification numbers.
2. Purchase Order (PO)	modules/purchases/orders.php Draft purchase order selecting supplier, items, required quantities, expected purchase rates, and GST.	Generates document PO-000001. Does not increment inventory yet (stock has not physically arrived).
3. Goods Received Note (GRN)	modules/purchases/grn.php Pharmacist verifies delivery boxes, checks batch numbers, manufacturing dates, and expiry dates. Records received quantities.	Inventory Inward Point: Creates new rows in medicine_batches. Writes PURCHASE_RECEIPT rows in pharmacy_stock_ledger. Stock is now sellable.
4. Purchase Invoice	modules/purchases/invoices.php Match distributor tax invoice with approved GRN items. Validates supplier bill number and GST amounts.	Generates PINV-000001. Updates supplier ledger balance with total accounts payable liability.
5. Supplier Payment	modules/purchases/payments.php Record payment voucher against outstanding invoices via Cheque, RTGS, NEFT, or Cash.	Generates SP-000001. Updates invoice status to PAID or PARTIALLY_PAID and reduces supplier outstanding.
6. Purchase Return	modules/purchases/returns.php Initiate return for cold-chain breaches, damaged ampoules, or short-expiry deliveries.	Generates Debit Note (PRT-000001), deducts inventory via PURCHASE_RETURN, and reduces accounts payable.

4. Inventory Management & FEFO Rules

A. The FEFO (First-Expiry, First-Out) Algorithm

Under the Drugs and Cosmetics Rules and NABH hospital accreditation guidelines, pharmacies must systematically dispense earliest-expiring stock first to minimize wastage and prevent administering expired medication.

System FEFO Allocation Logic:

1. Query all batches for the requested medicine where `quantity_available > 0` and `status = 'Active'`.
2. Filter out any batch where `expiry_date ≤ CURDATE()` (Expired).
3. Sort remaining batches by `expiry_date ASC, batch_id ASC`.
4. Fulfill requested units sequentially across batches until fully satisfied.

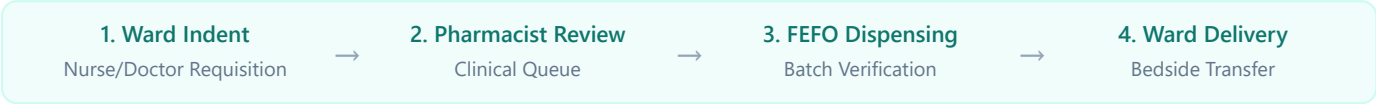
B. Inventory Operations & Safeguards

Module	Primary Function	Audit & Safety Controls
Stock Adjustments modules/inventory/adjustments.php	Physical cycle count discrepancy correction (+ or - units).	Requires reason code (Damage, Packaging defect, Count correction). Requires Admin authorization. Writes ADJUSTMENT_ADD or ADJUSTMENT_SUB to ledger.
Quarantine Isolation modules/inventory/quarantine.php	Isolate suspected, recalled, or near-expiry batches from billing.	Batches marked Quarantined are instantly blocked from OPD POS and IPD dispensing queues. Can be released or routed to disposal.
Expiry Management modules/inventory/expiry.php	Automated detection of expired and near-expiry (≤ 90 days) batches.	System automatically flags expired batches. Displays red warning badges. Prevents billing of expired units.
Disposal / Scrap modules/inventory/disposals.php	Formal destruction and discarding of expired or toxic substances.	Generates DISP-000001 destruction certificate recording destruction method (Incineration, Autoclave, Chemical treatment) and supervisor signature.
Stock Ledger modules/inventory/ledger.php	Complete chronological audit trail of every tablet, vial, and strip.	Searchable by medicine, batch, and date range. Guaranteed reconciliation with master stock.

5. Inpatient (IPD) Clinical Pharmacy & MAR

The Inpatient module handles the complete clinical medication cycle from hospital ward beds to bedside administration.

A. Ward Indenting & Clinical Dispensing



Sub-Module	Location	Operational Description
IPD Queue	modules/ipd/queue.php	Displays all active admitted hospital patients, current ward/bed assignments, and pending drug requests.
Ward Indents	modules/ipd/indents.php	Wards submit item requisitions. Pharmacists review dosage, stock availability, and approve order items.
Dispensing Record	modules/ipd/dispensing.php	Executes physical dispensing with batch verification. Generates DSP-000001 and updates patient ledger.

B. Medication Administration Record (MAR)

Located at: modules/ipd/mar.php (Sidebar: **Clinical MAR**)

Enables ward nursing staff to document the actual administration of scheduled doses at the patient's bedside, maintaining an airtight record for clinical safety and legal compliance.

- **Administered (GIVEN):** Dose successfully given to patient with time and nurse signature.
- **HELD:** Dose paused by doctor order (e.g. low blood pressure or fasting prior to surgery).
- **MISSED:** Dose missed due to clinical circumstance with mandatory justification reason.
- **REFUSED:** Patient declined medication. Recorded for physician review.

C. Discharge Medication Reconciliation

Located at: modules/ipd/reconciliation.php (Sidebar: **Reconciliation**)

Prior to patient discharge, the pharmacist performs a 3-way reconciliation comparing:

Indented Quantity ↔ Dispensed Quantity ↔ Administered Quantity (MAR)

Any unadministered medication is flagged for return and credit before the patient's final hospital invoice is cleared.

6. Reporting, Administration & Maintenance

A. Analytical Reports Suite

Report Name	Path	Primary Value & Analytics
Sales Analysis	modules/reports/sales.php	Channel breakdowns (OPD Sales vs IPD Ward Sales), daily sales registers, payment modes, and gross margins.
Stock Valuation	modules/reports/stock.php	Current inventory value at purchase rate and MRP, batch breakdown, and reorder alerts.
Procurement Register	modules/reports/purchases.php	Supplier-wise purchases, GST paid, payment statuses, and outstanding debt aging.
Expiry Forecast	modules/reports/expiry.php	Risk projection for batches expiring in 30, 60, and 90 days to prevent stock losses.
Fast / Slow Moving	modules/reports/fast_slow.php	ABC analysis of inventory velocity to optimize capital allocation.
MAR Register	modules/reports/mar_register.php	Clinical audit log of nursing administrations, missed doses, and adherence ratios.

B. System Settings & Document Sequences

Located at: modules/admin/settings.php and modules/admin/roles.php

- **Hospital Branding:** Manage Hospital Legal Name, Pharmacy License Number, GSTIN, and Hospital Logo.
- **Document Sequences:** Document sequence counters are managed authoritatively with atomic zero-padded numbering:
 - CS- (OPD Counter Sale Invoices)
 - RS- (IPD Regular Invoices)
 - PO- (Purchase Orders)
 - GRN- (Goods Received Notes)
 - PINV- (Purchase Invoices)
 - SRT- / PRT- (Sales & Purchase Return Debit/Credit Notes)
 - IND- (Ward Indents) & DSP- (Dispensing Records)

C. Routine Backup & Disaster Recovery

Recommended Daily Database Backup:

Run the following command via Windows Command Prompt / Task Scheduler to create an instant SQL backup:

```
"C:\xampp\mysql\bin\mysqldump.exe" -u root pharmacy_db > "C:\Backups\pharmacy_db_backup.sql"
```